

EX 10 :

Write a Python Program to Implement Expectation & Maximization Algorithm

PROGRAM :

```
from sklearn.datasets import load_iris
from sklearn.mixture import GaussianMixture
from sklearn.metrics import accuracy_score
import numpy as np
```

```
iris = load_iris()
```

```
X = iris.data
```

```
y = iris.target
```

```
# EM Algorithm using Gaussian Mixture Model
```

```
model = GaussianMixture(
    n_components=3,
    random_state=42
)
```

```
model.fit(X)
```

```
# Predict clusters
```

```
labels = model.predict(X)
```

```
print("Actual Classes:")
```

```
print(y)
```

```
print("\nPredicted Clusters:")
```

```
print(labels)
```

```
print("\nNumber of samples in each cluster:")
```

```
print(np.bincount(labels))
```

OUTPUT

```
.. Actual Classes:
[0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 2 2 2 2 2 2 2 2
 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
 2 2]

Predicted Clusters:
[1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
 1 1 1 1 1 1 1 1 1 1 1 1 1 2 2 2 2 2 2 2 2 2 2 2 2 2 2 0 2 0 2 0 2
 2 2 2 0 2 2 2 2 2 0 2 2 2 2 2 2 2 2 2 2 2 2 2 0 0 0 0 0 0 0 0 0
 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
 0 0]

Number of samples in each cluster:
[55 50 45]
```